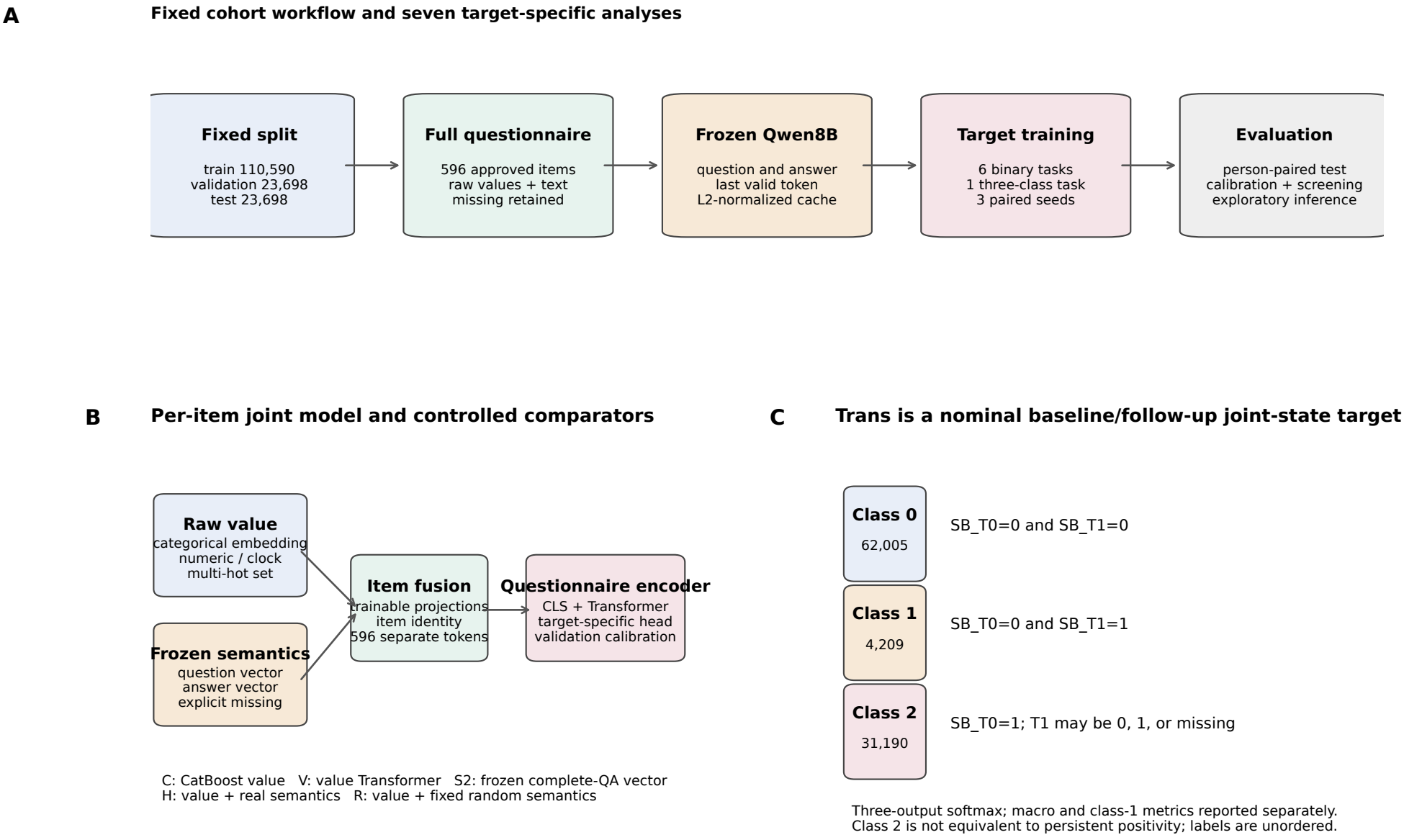
A
Fixed cohort workflow and seven target-specific analyses
Fixed split
train 110,590
validation 23,698
test 23,698
Full questionnaire
596 approved items
raw values + text
missing retained
Frozen Qwen8B
question and answer
last valid token
L2-normalized cache
Target training
6 binary tasks
1 three-class task
3 paired seeds
Evaluation
person-paired test
calibration + screening
exploratory inference
B
Per-item joint model and controlled comparators
Raw value
categorical embedding
numeric / clock
multi-hot set
Frozen semantics
question vector
answer vector
explicit missing
Item fusion
trainable projections
item identity
596 separate tokens
Questionnaire encoder
CLS + Transformer
target-specific head
validation calibration
C: CatBoost value V: value Transformer S2: frozen complete-QA vector
H: value + real semantics R: value + fixed random semantics
C
Trans is a nominal baseline/follow-up joint-state target
Class 0
62,005
SB_T0=0 and SB_T1=0
Class 1
4,209
SB_T0=0 and SB_T1=1
Class 2
31,190
SB_T0=1; T1 may be 0, 1, or missing
Three-output softmax; macro and class-1 metrics reported separately.
Class 2 is not equivalent to persistent positivity; labels are unordered.